Interior-Point Methods Strike Back: Solving the Wasserstein Barycenter Problem

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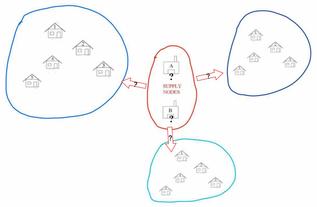
Wasserstein Barycenter Problem

Question: If we don't know which group to serve, How to decide the locations and manage the inventory?

For a set of probability distributions $\{\mathcal{P}^{(t)}, t = 1, 2, \cdots, N\}$ where $\mathcal{P}^{(t)} = \{(a_i^{(t)}, \boldsymbol{q}_i^{(t)}) : i = 1, \ldots, m_t\}$. The Wasserstein barycenter of those distributions is calculated as



$$\min_{\mathcal{P}} \frac{1}{N} \sum_{t=1}^{N} (\mathcal{W}_2(\mathcal{P}, \mathcal{P}^{(t)}))^2, \qquad (1)$$

where $\mathcal{W}_2(\cdot,\cdot)$ is the Wasserstain distance. For two measures $\mathcal{P}^{(t)} = \{(a_i^{(t)}, \boldsymbol{q}_i^{(t)}) : i = 1, \cdots, m_t\}$, $t = s,\ d$, the Wasserstein distance is

$$\mathcal{W}_2(\mathcal{P}^{(s)}, \mathcal{P}^{(d)}) := min \left\{ \sqrt{\sum_{i=1}^{m_s} \sum_{j=1}^{m_d} \pi_{ij} \|\boldsymbol{q}_i^{(s)} - \boldsymbol{q}_j^{(d)}\|^2} : \Pi = [\pi_{ij}] \in \mathcal{M}(\boldsymbol{a}^{(s)}, \boldsymbol{a}^{(d)}) \right\}, \qquad (2)$$

- Probability distributions: (locations and capacities) $\sum_{i=1}^{n} a_i^{(t)} = 1, t = s, d$.
- Measures with marginals $\boldsymbol{a}^{(s)}, \boldsymbol{d}^{(d)}$: (transportation plan) $\mathcal{M}(\boldsymbol{a}^{(s)}, \boldsymbol{a}^{(d)}) := \{\Pi \in \mathbb{R}_+^{m_s \times m_d} : \Pi \mathbf{1}_{m_s} = \boldsymbol{a}^{(s)}, \Pi^\top \mathbf{1}_{m_d} = \boldsymbol{a}^{(d)}\}$

Wasserstein Barycenter has many applications in

- Clustering
- Image processing
- Others

The right illustration is a comparison of algorithms for computing the barycenters between a Sinkhorn based approach[1](left) and MAAIPM(right).

Problem Reformulation

In general case, we can reformulate (1) as (3).

$$\begin{aligned} \min_{\boldsymbol{w}, X, \{\Pi^{(t)}\}} \quad & \textstyle\sum_{t=1}^{N} \left\langle \mathcal{D}(X, Q^{(t)}), \Pi^{(t)} \right\rangle \\ \text{s.t.} \quad & \Pi^{(t)} \mathbf{1}_{m_t} = \boldsymbol{w}, \left(\Pi^{(t)}\right)^\top \mathbf{1}_m = \boldsymbol{a}^{(t)}, \Pi^{(t)} \geq 0, \forall t \mathbf{1}_m^\top \boldsymbol{w} = 1, \boldsymbol{w} \geq 0 \end{aligned} \qquad (3)$$

- $\langle X, Q \rangle = \sum_{ij} X_{ij} Q_{ij}$
- Weight: $\boldsymbol{w} := (w_1, \cdots, w_m)^\top \in R_+^m$,
- Support points: $X := [\boldsymbol{x}_1, \cdots, \boldsymbol{x}_m] \in \mathbb{R}^{d \times m}$
- Measures with marginals: $\Pi^{(t)} \in R_+^{m \times m_t}$ for $t = 1, \cdots, N$
- Euclidean distances: $\mathcal{D}(X, Q^{(t)}) := [\|\boldsymbol{x}_i - \boldsymbol{q}_j^{(t)}\|^2] \in \mathbb{R}^{m \times m_t}$ for $t = 1, \cdots, N$

Remark: With w and Π fixed, the objective value of original problem (3) is quadratic in terms of X, and the optimal X is solved by:

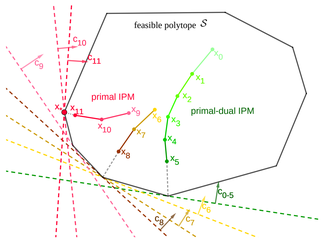
$$\boldsymbol{x}_i := \frac{1}{N w_i} \sum_{t=1}^{N} \sum_{j=1}^{m_t} \pi_{ij}^{(t)} \boldsymbol{q}_j^{(t)} \qquad (4)$$

In another word, the original problem(3) can be reformulated as

$$\min \boldsymbol{c}(\boldsymbol{x})^\top \boldsymbol{x} \ \text{ s.t.} A\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq 0. \qquad (5)$$

Matrix-based Adaptive Alternating IPM (MAAIPM)

a) x_i is returned by each IPM iteration under the **objective gradient** c_i; and c_i is calculated by x_{i-1};
b) At the beginning, c_i is updated after every a few **primal-dual IPM** iterations (green).
c) Then **primal IPM** (yellow and red) is applied to frequently update c and uses "jump" tricks to escape local minima.



Solve the Normal Equations

In presolve phase, we get $\bar{A}$ whose all rows are linear independant. No matter which IPM we use, the main computational cost lies in solving the Newton equations

$$\bar{A}(D^k)^2 \bar{A}^\top \Delta \boldsymbol{\lambda}^k = \boldsymbol{f}^k, \qquad (6)$$

Perform several basic transformations on the matrix $\bar{A} D \bar{A}^T$

$$V \bar{A} D \bar{A}^T V^\top = \begin{bmatrix} B_1 & & \\ & A_1 + A_2 & \\ & & c \end{bmatrix},$$

in which $A_2 = \left(\mathbf{1}_N \mathbf{1}_N^\top\right) \otimes (R^\top R)$, $A_1 = diag(A_{11}, A_{22}, \cdots, A_{nn})$, and B_1 is a diagnal matrix. Use low-rank regularization:

$$(A_1 + A_2)^{-1} = A_1^{-1} - A_1^{-1} U \bar{R} (I + \bar{R}^\top U^\top A_1^{-1} U \bar{R})^{-1} \bar{R}^\top U^\top A_1^{-1},$$

where

$$\bar{R} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ R^\top \end{bmatrix}, \quad U = \begin{bmatrix} I_{m-1} & & I_{m-1} \\ & I_{m-1} & \vdots \\ & & \ddots\ I_{m-1} \\ & & I_{m-1} \end{bmatrix}.$$

$$\bar{R}(I + \bar{R}^\top U^\top A_1^{-1} U \bar{R})^{-1} \bar{R}^\top = (\boldsymbol{e}_N \boldsymbol{e}_N^\top) \otimes \left(R^\top \left(I + R(\sum_{i=1}^{N} A_{ii}^{-1}) R^\top\right)^{-1} R \right),$$

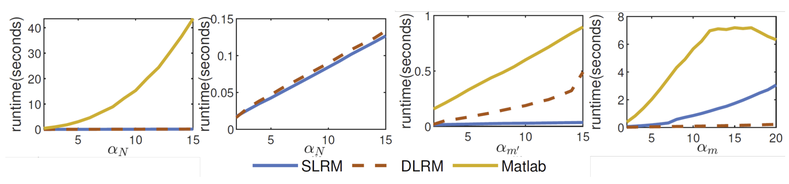
and $\mathbf{e}_N = (0, 0, \cdots, 1)^\top$.

Theoretical Results

- Suitable for **parallel computing**
- When $m = m_t$, the complexity is decreased from $O(N^3 m^4)$ to $O(N m^3)$
- When the support points are different, the **memory usage** is within a constant multiple of that of the first order Sinkhorn based method IBP[1]

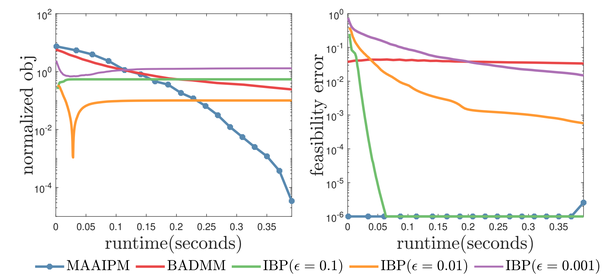
  Single Low-rand regularization is linear with respect to N and m_t:
  Time complexity $O(m^2 \sum m_i + N m^3)$, memory usage $O(m \sum m_i + N m^2)$

  Double Low-rand regularization is linear with respect to N and m:
  Time complexity $O(m \sum m_i^2 + \sum m_i^3)$, memory usage $O(m \sum m_i + \sum m_i^2)$

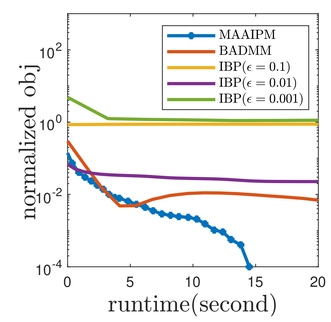


Experiments of Wasserstein Barycenter Problems

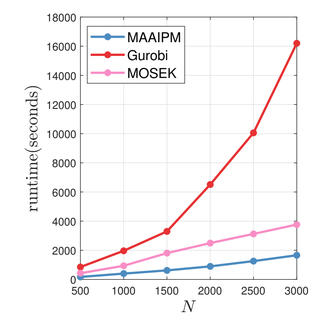
Pre-Specified Support Case



Free Support Case



Low Memory Usage



Comparison with IBP[1] and BADMM[5] on Datasets

	MNIST			Fashion-MNIST		
time(seconds)	250	500	1000	25	50	75
MAAIPM						
BADMM						
IBP($\epsilon = 0.01$)						

When the support points of distributions are different, the convolutional Sinkorn based method[3] is no longer applicable

Acknowledgement & References

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